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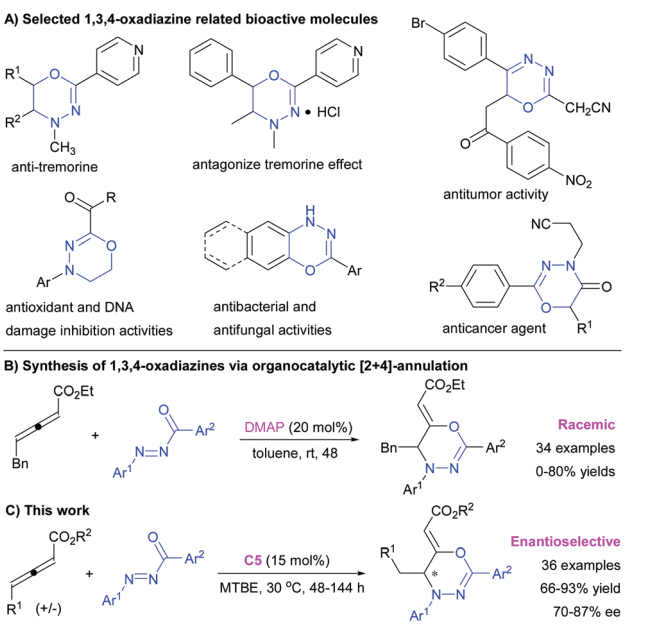
# Organocatalytic enantioselective [2 + 4]-annulation of γ-substituted allenoates with *N*-acyldiazenes for the synthesis of optically active 1,3,4-oxadiazines†

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**An enantioselective [2 + 4]-annulation of γ-substituted allenoates with *N*-acyldiazenes has been developed for the first time. In the presence of an L-proline-derived DMAP analogue, the annulation proceeded smoothly to afford a broad range of 1,3,4-oxadiazine derivatives in good to excellent yields with high stereoselectivities.**

1,3,4-Oxadiazine is a class of core structural units widely occurring in a number of bioactive molecules, which exhibit diversified bioactivities (Scheme 1A).[1] Accordingly, the efficient synthesis of these heterocycles featuring 1,3,4-oxadiazine has therefore attracted intensive interest.[2] In terms of organocatalysis, Wang and Meng *et al.* reported a 4-dimethylaminopyridine (DMAP)-catalyzed [2 + 4]-cycloaddition between allenoates[3] and *N*-acyldiazenes[4] for the synthesis of racemic 1,3,4-oxadiazine derivatives (Scheme 1B).[5] It was proposed that the cycloaddition is triggered by the $S_N2$ attack of nucleophilic DMAP on the allenoate to give a zwitterionic intermediate, followed by γ-addition of *N*-acyldiazene, intramolecular Michael addition of an oxygen anion, and elimination of DMAP to regenerate the catalyst and afford the desired product. The density functional theory calculation indicated that the addition of DMAP to γ-methyl allenoate is the rate-determining step, and the reaction of chiral γ-methyl allenoate and *N*-acyldiazene is both the regio- and enantioselective-determining step.[6] It should be noted that many organocatalytic asymmetric reactions of γ-substituted allenoates[7] and *N*-acyldiazenes[8] have been established previously, but the organocatalytic stereoselective [2 + 4]-annulation of racemic γ-substituted allenoates and *N*-acyldiazenes has not been realized yet. The development of an organocatalytic asymmetric synthesis strategy for optically active 1,3,4-oxadiazines is therefore highly desirable.

As nucleophilic catalysts, enantioenriched structurally diverse DMAP analogues have been particularly attractive targets in the field of asymmetric organocatalysis.[9] For examples, L-proline-derived DMAP analogues have been successfully employed as efficient catalysts in the asymmetric kinetic resolution of *sec*-alcohols,[10] and [3 + 2]-annulation of MBH carbonates.[11] In consideration of a similar activation mechanism with MBH carbonate, allenoates were expected to be activated by L-proline-derived DMAP analogues to generate the desired enantioenriched products. Based on our previous works on organocatalytic asymmetric annulations of MBH carbonates,[12] and as a part of our ongoing interests in the chem-




**Scheme 1** Related 1,3,4-oxadiazines and synthetic strategies.

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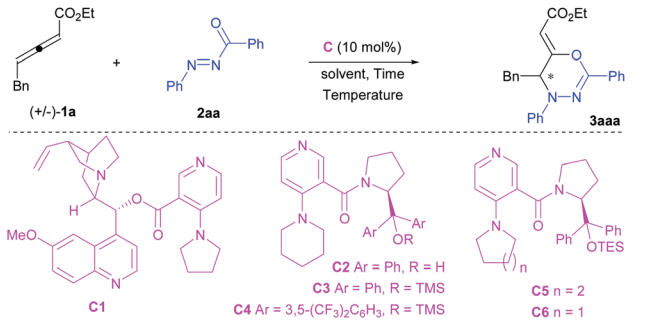
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istry of allene,[13] herein we report an L-proline-derived DMAP analogue catalyzed [2 + 4]-annulation between γ-substituted allenoates and *N*-acyldiazenes towards optically active 1,3,4-oxadiazine derivatives (Scheme 1C).

To address the enantioselectivity issue, the annulation of racemic γ-benzyl allenoate **1a** with *N*-phenyl-*N*′-benzoyldiazene **2aa** was investigated as a model reaction (Table 1). Inspired by Wang and Meng' work, we initially focused on the catalyst screening of chiral DMAP analogues. In the presence of catalyst **C1** derived from quinine, a trace of the desired product **3aaa** was obtained (Table 1, entry 1). An essential enhancement was achieved when the L-proline-derived DMAP analogue **C2** was employed as the catalyst, affording the desired product **3aaa** in 32% yield with 29% ee (Table 1, entry 2). A further survey on L-proline-derived DMAP analogues indicated that all probed catalysts could prompt the reaction to afford the desired 1,3,4-oxadiazine **3aaa** with fluctuating enantioselectivities and yields (Table 1, entries 1–6). In particular, catalyst **C5** enabled the formation of the desired product **3aaa** in 32% yield with 75% ee (Table 1, entry 5). To obtain a satisfactory yield and stereoselectivity, further optimization of reaction conditions was then carried out in the presence of catalyst **C5**. The screening of reaction media indicated that the solvent had a significant impact on the yield and a slight effect on the enantioselectivity (Table 1, entries 7–14). Notably, the yield was improved to 67% without compromising the ee value when the **C5** catalyzed reaction was carried out in methyl *t*-butyl ether (MTBE, Table 1, entry 14). It was found that both yield and ee were sensitive to the concentration (Table 1, entries 15–17), and 1.5 mL MTBE was identified to be more suitable to afford product **3aaa** in 65% yield with 75% ee. No further improvement in terms of enantioselectivity and yield was obtained after the evaluation of the reactant ratio (Table 1, entries 18–20). Pleasingly, the desired 1,3,4-oxadiazine **3aaa** could be obtained in 70% yield with 74% ee when the reaction was carried out at 30 °C (Table 1, entry 21). Further modifying the catalyst loading and reaction time revealed that product **3aaa** could be formed in 82% yield with 71% ee (Table 1, entries 23–25). As a result, the optimal reaction conditions were identified as follows: *N*-phenyl-*N*′-benzoyldiazene **2aa** (0.3 mmol) was added to a solution of racemic γ-benzyl allenoate **1a** (0.2 mmol) with **C5** (15 mol%) in MTBE (1.5 mL) at 30 °C and reacted for 48 h; the desired 1,3,4-oxadiazine **3aaa** was obtained in 82% with 71% ee.

Having identified the optimal conditions, we sought to explore the scope of the **C5**-catalyzed [2 + 4]-annulation between γ-substituted allenoates **1** and *N*-acyldiazenes **2**. As shown in Table 2, we first examined the scope of *N*-acyldiazenes **2** by the **C5**-catalyzed [2 + 4]-annulation of racemic γ-benzyl allenoate **1a**. Generally, these reactions proceeded smoothly to afford the corresponding 1,3,4-oxadiazines **3** with good yields and enantioselectivity. Importantly, various substituents, including electron-withdrawing (F, Cl, Br) and electron-donating (Me) groups, could be introduced to the *para*-position of the aromatic ring of $Ar^2$ with only minor effects on the yields and asymmetric induction (Table 2, entries 1–5). Notably, the **C5**-catalyzed [2 + 4]-annulation of 2-methyl-*N*-(phenylimino)benzamide **2af** also afforded the desired 1,3,4-oxadiazine **3aaf** in 88% yield with 71% ee (Table 2, entry 6). Accordingly, the reactions of the bulk *N*-acyldiazenes **2** were investigated. 1-Naphthyl ($Ar^2$) was found to be tolerated, furnishing the product **3aag** in 77% yield with 70% ee (Table 2, entry 7). Using *N*-acyldiazene **2ah** ($Ar^2$ = 3,4-dimethoxylphenyl) enabled the formation of the product **3aah** in 71% yield with 74% ee (Table 2, entry 8). Similar results were obtained from the reaction of *N*-acyldiazene **2ai** ($Ar^2$ = 3,5-dimethoxylphenyl, Table 2, entry 9). In particular, 1,3,4-oxadiazine **3aaj** was obtained in 77% yield with 78% ee from the **C5**-catalyzed [2 + 4]-annulation of *N*-acyldiazene **2aj** ($Ar^2$ = 3,4,5-trimethoxylphenyl, Table 2, entry 10). The sterically encumbered *N*-acyldiazene **2ak** ($Ar^2$ = 1-adamantyl) was also compatible to furnish the corresponding product **3aak** in 87%

**Table 1** Condition optimization



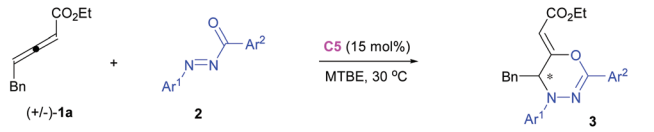

| Entry[a] | C | Solvent | *T* [°C] | Time [h] | Yield[b] [%] | ee[c] |
|---|---|---|---|---|---|---|
| 1 | **C1** | Toluene | rt | 48 | **3aaa**, trace | — |
| 2 | **C2** | Toluene | rt | 48 | **3aaa**, 32 | 29 |
| 3 | **C3** | Toluene | rt | 48 | **3aaa**, 23 | 68 |
| 4 | **C4** | Toluene | rt | 48 | **3aaa**, 6 | 35 |
| 5 | **C5** | Toluene | rt | 48 | **3aaa**, 32 | 75 |
| 6 | **C6** | Toluene | rt | 48 | **3aaa**, 29 | 68 |
| 7 | **C5** | EtOAc | rt | 48 | **3aaa**, 66 | 69 |
| 8 | **C5** | MeCN | rt | 48 | **3aaa**, 59 | 47 |
| 9 | **C5** | $CH_2Cl_2$ | rt | 48 | **3aaa**, 35 | 70 |
| 10 | **C5** | $CHCl_3$ | rt | 48 | **3aaa**, 21 | 71 |
| 11 | **C5** | $Et_2O$ | rt | 48 | **3aaa**, 39 | 70 |
| 12 | **C5** | THF | rt | 48 | **3aaa**, 57 | 69 |
| 13 | **C5** | 1,4-Dioxane | rt | 48 | **3aaa**, 66 | 73 |
| 14[d] | **C5** | MTBE | rt | 48 | **3aaa**, 67 | 74 |
| 15 | **C5** | MTBE (1.5 mL) | rt | 48 | **3aaa**, 65 | 75 |
| 16 | **C5** | MTBE (1.0 mL) | rt | 48 | **3aaa**, 57 | 76 |
| 17 | **C5** | MTBE (3.0 mL) | rt | 48 | **3aaa**, 77 | 71 |
| 18[e] | **C5** | MTBE (1.5 mL) | rt | 48 | **3aaa**, 77 | 71 |
| 19[f] | **C5** | MTBE (1.5 mL) | rt | 48 | **3aaa**, 70 | 72 |
| 20[g] | **C5** | MTBE (1.5 mL) | rt | 48 | **3aaa**, 79 | 69 |
| 21 | **C5** | MTBE (1.5 mL) | 30 | 48 | **3aaa**, 70 | 74 |
| 22 | **C5** | MTBE (1.5 mL) | 35 | 48 | **3aaa**, 70 | 71 |
| 23[h] | **C5** | MTBE (1.5 mL) | 30 | 42 | **3aaa**, 74 | 72 |
| 24[h] | **C5** | MTBE (1.5 mL) | 30 | 48 | **3aaa**, 82 | 71 |
| 25[h] | **C5** | MTBE (1.5 mL) | 30 | 54 | **3aaa**, 74 | 71 |

[a] Unless noted, a mixture of (±)-**1a** (0.3 mmol), **2aa** (0.2 mmol) and **C** (10 mol%) in the solvent (2.0 mL) at the indicated temperature for the time given. [b] Isolated yield. [c] Determined by chiral HPLC analysis. [d] MTBE = methyl *t*-butyl ether. [e] (±)-**1a** (0.4 mmol). [f] (±)-**1a** (0.2 mmol), **2aa** (0.3 mmol). [g] (±)-**1a** (0.3 mmol), **2aa** (0.4 mmol). [h] **C5** (15 mol%).


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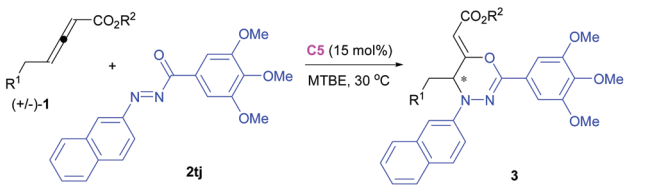
**Table 2** Scope of *N*-acyldiazenes **2**




| Entry[a] | $Ar^1/Ar^2$ | Time [h] | Yield[b] [%] | ee[c] |
|---|---|---|---|---|
| 1 | Ph/Ph | 48 | **3aaa**, 82 | 71 |
| 2 | Ph/4-$FC_6H_4$ | 48 | **3aab**, 72 | 72 |
| 3 | Ph/4-$ClC_6H_4$ | 48 | **3aac**, 73 | 71 |
| 4 | Ph/4-$BrC_6H_4$ | 60 | **3aad**, 66 | 70 |
| 5 | Ph/4-$MeC_6H_4$ | 48 | **3aae**, 76 | 70 |
| 6 | Ph/2-$MeC_6H_4$ | 48 | **3aaf**, 88 | 71 |
| 7 | Ph/1-naphthyl | 60 | **3aag**, 77 | 70 |
| 8 | Ph/3,4-$(MeO)_2C_6H_3$ | 60 | **3aah**, 71 | 74 |
| 9 | Ph/3,5-$(MeO)_2C_6H_3$ | 72 | **3aai**, 72 | 72 |
| 10 | Ph/3,4,5-$(MeO)_3C_6H_2$ | 60 | **3aaj**, 77 | 78 |
| 11 | Ph/1-adamantyl | 60 | **3aak**, 87 | 75 |
| 12 | 2-$FC_6H_4$/3,4,5-$(MeO)_3C_6H_2$ | 60 | **3abj**, 82 | 75 |
| 13 | 2-$ClC_6H_4$/3,4,5-$(MeO)_3C_6H_2$ | 60 | **3acj**, 84 | 78 |
| 14 | 2-$BrC_6H_4$/3,4,5-$(MeO)_3C_6H_2$ | 60 | **3adj**, 81 | 76 |
| 15 | 2-$MeC_6H_4$/3,4,5-$(MeO)_3C_6H_2$ | 60 | **3aej**, 81 | 78 |
| 16 | 3-$ClC_6H_4$/3,4,5-$(MeO)_3C_6H_2$ | 60 | **3afj**, 76 | 74 |
| 17 | 3-$BrC_6H_4$/3,4,5-$(MeO)_3C_6H_2$ | 60 | **3agj**, 81 | 75 |
| 18 | 3-$MeC_6H_4$/3,4,5-$(MeO)_3C_6H_2$ | 60 | **3ahj**, 84 | 77 |
| 19 | 4-$FC_6H_4$/3,4,5-$(MeO)_3C_6H_2$ | 60 | **3aij**, 83 | 79 |
| 20 | 4-$BrC_6H_4$/3,4,5-$(MeO)_3C_6H_2$ | 60 | **3ajj**, 77 | 80 |
| 21 | 4-$IC_6H_4$/3,4,5-$(MeO)_3C_6H_2$ | 60 | **3akj**, 78 | 78 |
| 22 | 4-$CF_3C_6H_4$/3,4,5-$(MeO)_3C_6H_2$ | 60 | **3alj**, 75 | 72 |
| 23 | 4-i-$PrC_6H_4$/3,4,5-$(MeO)_3C_6H_2$ | 60 | **3amj**, 72 | 73 |
| 24 | 4-*t*-$BuC_6H_4$/3,4,5-$(MeO)_3C_6H_2$ | 60 | **3anj**, 76 | 72 |
| 25 | 4-$BnOC_6H_4$/3,4,5-$(MeO)_3C_6H_2$ | 60 | **3aoj**, 85 | 73 |
| 26 | 2,4-$F_2C_6H_3$/3,4,5-$(MeO)_3C_6H_2$ | 60 | **3apj**, 92 | 75 |
| 27 | 2,4-$Cl_2C_6H_3$/3,4,5-$(MeO)_3C_6H_2$ | 60 | **3aqj**, 93 | 75 |
| 28 | 2,5-$Cl_2C_6H_3$/3,4,5-$(MeO)_3C_6H_2$ | 60 | **3arj**, 77 | 74 |
| 29 | 3,4-$Me_2C_6H_3$/3,4,5-$(MeO)_3C_6H_2$ | 60 | **3asj**, 87 | 76 |
| 30 | 2-Naphthyl/3,4,5-$(MeO)_3C_6H_2$ | 60 | **3atj**, 76 | 87 |
| 31 | 2-Pyridyl/3,4,5-$(MeO)_3C_6H_2$ | 60 | **3auj**, 74 | 82 |

[a] Unless noted, a mixture of (±)-**1a** (0.3 mmol), **2** (0.2 mmol) and **C5** (15 mol%) in MTBE (1.5 mL) at 30 °C for the time given. [b] Isolated yield. [c] Determined by chiral HPLC analysis.

**Table 3** Scope of γ-substituted allenoates **1**




| Entry[a] | $R^1/R^2$ | Time [h] | Yield[b] [%] | ee[c] |
|---|---|---|---|---|
| 1 | Me/Et | 60 | **3btj**, 76 | 75 |
| 2 | Et/Et | 60 | **3ctj**, 73 | 82 |
| 3 | Ph/*t*-Bu | 144 | **3dtj**, 72 | 76 |
| 4 | Ph/Bn | 60 | **3etj**, 81 | 81 |
| 5 | Ph/$CHPh_2$ | 60 | **3ftj**, 82 | 74 |

[a] Unless noted, a mixture of (±)-**1** (0.3 mmol), **2tj** (0.2 mmol) and **C5** (15 mol%) in MTBE (1.5 mL) at 30 °C for the time given. [b] Isolated yield. [c] Determined by chiral HPLC analysis.

yield with 75% ee (Table 2, entry 11). With these encouraging results in hand, the effect of the aromatic residue $Ar^1$ on the **C5**-catalyzed [2 + 4]-annulation was then examined. Pleasingly, both electron-withdrawing (F, Cl, Br, I, $CF_3$) and electron-donating (Me, i-Pr, *t*-Bu, BnO) groups on the aromatic ring of $Ar^1$ were accommodated under the standard conditions to furnish products **3abj–asj** in 72–93% yields with 72–80% ee (Table 2, entries 12–29). Furthermore, the reaction of *N*-acyldiazene **2tj** ($Ar^1$ = 2-naphthyl, $Ar^2$ = 3,4,5-trimethoxylphenyl) generated the corresponding product **3atj** in 76% yield with 87% ee (Table 2, entry 30). In addition, the heteroaromatic *N*-acyldiazene **2uj** ($Ar^1$ = 2-pyridinyl) also reacted smoothly with γ-benzyl allenoate **1a** to afford the product **3auj** in 74% yield with 82% ee (Table 2, entry 31). Taken altogether, the **C5**-catalyzed [2 + 4]-annulation of γ-benzyl allenoate **1a** was successfully extended to a broad scope of *N*-acyldiazenes, furnishing the desired 1,3,4-oxadiazine derivatives in good to excellent yields with high stereoselectivities.

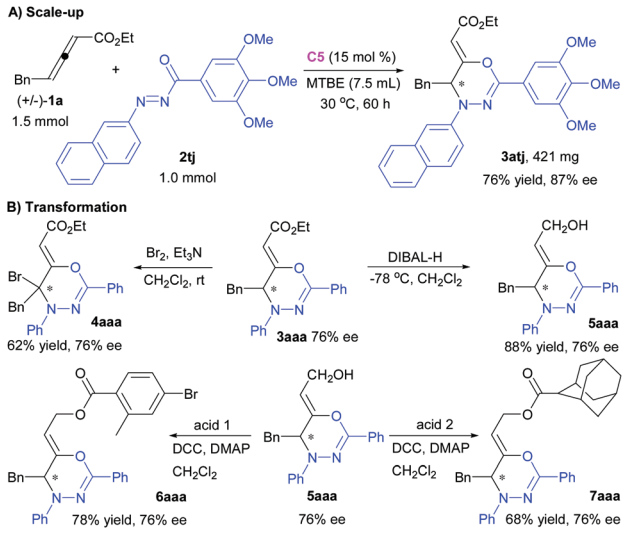
To further explore the scope and limitations of the reaction, the scope of γ-substituted allenoates **1** was then examined by the **C5**-catalyzed [2 + 4]-annulation of *N*-acyldiazene **2tj** (Table 3). Under the standard conditions, γ-ethyl allenoate **1b** reacted smoothly with *N*-acyldiazene **2tj**, affording the corresponding product **3btj** in 76% yield with 75% ee (Table 3, entry 1). The reaction of γ-propyl allenoate **1c** enabled the formation of product **3ctj** in 73% yield with 82% ee (Table 3, entry 2). The ester group ($CO_2R^2$) of allenoates **1** was found to affect the reaction slightly, furnishing 1,3,4-oxadiazines **3dtj–ftj** in 72–82% yield with 74–81% ee (Table 3, entries 3–5). In all, an organocatalytic enantioselective [2 + 4]-annulation of racemic γ-substituted allenoates and *N*-acyldiazenes has been established, which featured mild conditions, wide substrate scope, high efficiency and stereoselectivity. Importantly, this method provided a direct way to optically activate 1,3,4-oxadiazines.

With these encouraging results in hand, we then tried to determine the absolute configuration of products **3** by X-ray crystallography. Interestingly, the crystal of product **3aac** was found to contain a pair of enantiomers (*R*)-**3aac** and (*S*)-**3aac**.[14a] The ee value of the residual product **3aac** in the mother solution turned to be 99%. Furthermore, the crystal of product **3aak** was also comprised of racemic **3aak**.[14b]

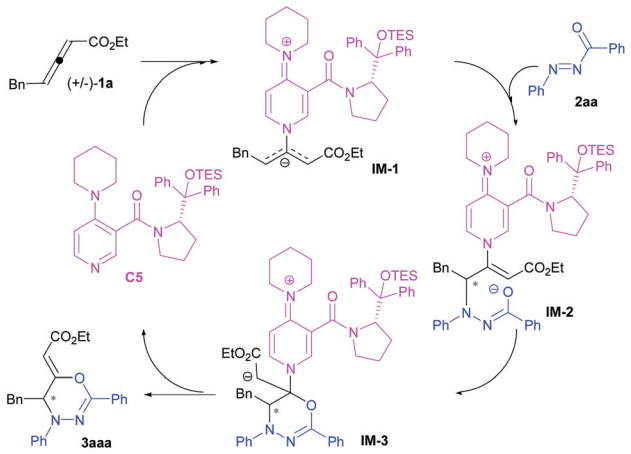
To illustrate the synthetic potential of this reaction, a scale-up experiment was carried out under the optimal reaction conditions. The **C5**-catalyzed [2 + 4]-annulation between 1.5 mmol of racemic γ-benzyl allenoate **1a** and 1.0 mmol of *N*-acyldiazene **2tj** proceeded smoothly to furnish the desired 1,3,4-oxadiazine **3atj** in 76% yield with 87% ee (Scheme 2A). Upon treatment with $Br_2/Et_3N$ in dichloromethane at room temperature, bromination of the product **3aaa** proceeded smoothly to furnish the corresponding product **4aaa** in 62% yield with 76% ee (Scheme 2B). Reduction of the ester group afforded alcohol **5aaa** in 88% yield without compromising the enantioselectivity. Further esterification of chiral alcohol **5aaa** generated products **6aaa** in 78% yield with 76% ee and **7aaa** in 68% yield with 76% ee, respectively.

According to the literature[5] and based on our results, a probable reaction mechanism was proposed in Scheme 3. An

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Scheme 2 Further investigations.




Scheme 3 The proposed mechanism.

initial addition of nucleophilic L-proline-derived DMAP analogue **C5** on the racemic γ-benzyl allenoate **1a** delivered the intermediate **IM-1**. Then the conjugate addition of **IM-1** to *N*-acyldiazene **2aa** furnished the intermediate **IM-2**, followed by an intramolecular 1,4-addition to afford the intermediate **IM-3**. Finally, the elimination reaction led to the formation of the desired product **3aaa** and regeneration of the chiral catalyst **C5**. Obviously, both the regio- and enantioselectivity originated from the reaction of the chiral intermediate **IM-1** with *N*-acyldiazene **2aa**.

In conclusion, we have successfully developed an efficient organocatalytic asymmetric [2 + 4] annulation of γ-substituted allenoates with *N*-acyldiazenes. In the presence of an L-proline-derived DMAP analogue, the reaction proceeds well to furnish a broad range of chiral 1,3,4-oxadiazines with high yields and stereoselectivities.

## Conflicts of interest

There are no conflicts to declare.

## Acknowledgements

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